Light

GCSE Computer Science — Boolean Logic (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Boolean Logic

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A smartphone unlocks when the correct fingerprint is scanned AND the screen is touched.
What type of logic operation does this describe?

- ○ **A.** OR
- ○ **B.** NOT
- ○ **C.** AND
- ○ **D.** IF

Q2 (1 mark)

A cinema offers discounts to customers who are students OR senior citizens.
What type of logic operation determines if a customer gets a discount?

- ○ **A.** AND
- ○ **B.** OR
- ○ **C.** NOT
- ○ **D.** MINUS

Q3 (1 mark)

A dishwasher runs when the door is NOT open.
The door is currently open (open = 1).
What is the output for NOT open?

- ○ **A.** 0 (dishwasher does not run)
- ○ **B.** 1 (dishwasher runs)
- ○ **C.** 2 (dishwasher runs slowly)
- ○ **D.** Open

Q4 (1 mark)

An OR gate has inputs H = 0 and I = 0.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** HI

Q5 (1 mark)

An AND gate has inputs U = 0 and V = 0.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** UV
- ○ **D.** Undefined

Q6 (1 mark)

What are the two possible outputs of a NOT gate?

- ○ **A.** 0 and 0
- ○ **B.** 1 and 1
- ○ **C.** 0 and 1
- ○ **D.** 0 and 2

Q7 (1 mark)

A logic circuit has 2 inputs: U and V.
How many rows are needed in the truth table?

- ○ **A.** 2 rows
- ○ **B.** 4 rows
- ○ **C.** 6 rows

- ○ **D.** 8 rows

Q8 (1 mark)

A printer prints when paper is loaded AND ink is available.
Paper is loaded but ink is empty.
Will the printer print?

- ○ **A.** Yes, because paper is loaded
- ○ **B.** Yes, because one condition is met
- ○ **C.** No, because both conditions must be met
- ○ **D.** No, because printers always need maintenance

Q9 (1 mark)

An arcade game gives a bonus life when the player collects 100 coins OR defeats a boss.
A player defeats a boss but has only 50 coins.
Does the player get a bonus life?

- ○ **A.** No, because 100 coins are needed
- ○ **B.** No, because both conditions must be met
- ○ **C.** Yes, because at least one condition is met
- ○ **D.** Yes, because 50 coins is close enough

Q10 (1 mark)

Which notation represents the NOT operation in Boolean algebra?

- A. ∧
- B. ∨
- C. ¬ or a bar over the variable
- D. +

Q11 (1 mark)

In Boolean logic, what are the only two possible values?

- A. Yes and No
- B. 0 and 1
- C. True and Maybe
- D. High and Medium

Q12 (1 mark)

An AND gate and an OR gate both have 2 inputs.
How many inputs does a NOT gate have?

- A. 0 inputs
- B. 1 input
- C. 2 inputs
- D. 3 inputs

Q13 (2 marks)

Evaluate the Boolean expression: H OR (NOT I)
Where H = 0 and I = 1.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** H
- ○ **D.** I

Q14 (2 marks)

Evaluate the Boolean expression: (U AND V) OR W
Where U = 1, V = 1, W = 0.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** UVW
- ○ **D.** Undefined

Q15 (2 marks)

In a 3-input truth table with inputs H, I, J, the rows are listed in binary counting order.
The first row is H=0, I=0, J=0.
What are the inputs in the fourth row?

- ○ **A.** H=0, I=1, J=0
- ○ **B.** H=0, I=1, J=1

- **C.** H=1, I=0, J=0
- **D.** H=1, I=1, J=1

Q16 (2 marks)

Here is part of a truth table:

H	I	Output
0	0	1
0	1	?
1	0	0
1	1	0

The circuit implements NOT H.
What value replaces the question mark?

- **A.** 0
- **B.** 1
- **C.** H
- **D.** I

Q17 (2 marks)

A 2-input logic gate has this truth table:

U	V	Output
0	0	0

0	1	1
1	0	1
1	1	1

What type of gate is this?

- ○ **A.** AND gate
- ○ **B.** OR gate
- ○ **C.** NOT gate
- ○ **D.** BUFFER gate

Q18 (2 marks)

A burglar alarm sounds when:
Motion is detected AND (Door sensor triggered OR Window sensor triggered)
Motion is detected, no door triggered, but window is triggered.
Will the alarm sound?

- ○ **A.** No, because door is not triggered
- ○ **B.** No, because not all sensors are triggered
- ○ **C.** Yes, because motion AND window are both triggered
- ○ **D.** Yes, because at least one sensor is triggered

Q19 (2 marks)

Which two Boolean expressions are equivalent?

- **A.** A AND B, and A OR B
- **B.** NOT (NOT A), and A
- **C.** A AND B, and B AND A
- **D.** Both second and third options are correct

Q20 (2 marks)

A NOT gate has this truth table:

Input	Output
0	1
1	0

What pattern describes the relationship between input and output?

- **A.** Output equals input
- **B.** Output is always 1
- **C.** Output is always 0
- **D.** Output is the opposite of input

Q21 (2 marks)

Evaluate: NOT (H AND I)
Where H = 1 and I = 0.

- **A.** 0
- **B.** 1

- ○ **C.** H
- ○ **D.** I

Q22 (2 marks)

A washing machine starts when:
Door is closed AND (Start button pressed OR Timer activated)
The door is open, start button is pressed, and timer is activated.
Will the machine start?

- ○ **A.** Yes, because start button is pressed
- ○ **B.** Yes, because timer is activated
- ○ **C.** No, because the door must be closed
- ○ **D.** No, because all three conditions must be met

Q23 (2 marks)

Evaluate: U AND 1
Where U = 1.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** U

Q24 (3 marks)

Complete the truth table for: (H AND I) OR J

H	I	J	Output
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	?
1	1	1	1

What value replaces the question mark?

- **A.** 0
- **B.** 1
- **C.** H
- **D.** HIJ

Q25 (3 marks)

A circuit implements: (NOT U) AND (NOT V)
For which input combination is the output 1?

- **A.** U=0, V=0
- **B.** U=0, V=1
- **C.** U=1, V=0

○ **D.** U=1, V=1

Q26 (3 marks)

A drone can fly when:
• Battery above 20% AND GPS signal present, OR
• Manual control mode is active
Which Boolean expression correctly represents this?

○ **A.** Battery AND GPS AND Manual

○ **B.** Battery OR GPS OR Manual

○ **C.** (Battery AND GPS) OR Manual

○ **D.** (Battery OR GPS) AND Manual

Q27 (3 marks)

A logic circuit has this truth table:

U	V	Output
0	0	0
0	1	1
1	0	1
1	1	0

Which Boolean expression matches this truth table?

○ **A.** U AND V

○ **B.** U OR V

- **C.** (U OR V) AND NOT (U AND V)
- **D.** (U AND V) OR NOT (U OR V)

Q28 (3 marks)

Evaluate: (NOT H) AND (I OR J)
Where H = 0, I = 0, J = 1.

- **A.** 0
- **B.** 1
- **C.** HIJ
- **D.** NOT

Q29 (3 marks)

A smart thermostat turns on heating when:
(Temperature LOW AND NOT Summer) OR (Emergency heat mode AND Battery backup)
Currently: Temperature is LOW, it is NOT Summer, Emergency mode is OFF, Battery is ON.
Will the heating turn on?

- **A.** No, because Emergency mode is off
- **B.** No, because both conditions in the expression must be true
- **C.** Yes, because Temperature is LOW and it is NOT Summer
- **D.** Yes, because Battery is on

Q30 (3 marks)

A logic circuit implements: (H OR I) AND J
In a complete 3-input truth table, how many rows have an output of 1?

- ○ **A.** 1 row
- ○ **B.** 2 rows
- ○ **C.** 3 rows
- ○ **D.** 4 rows

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.